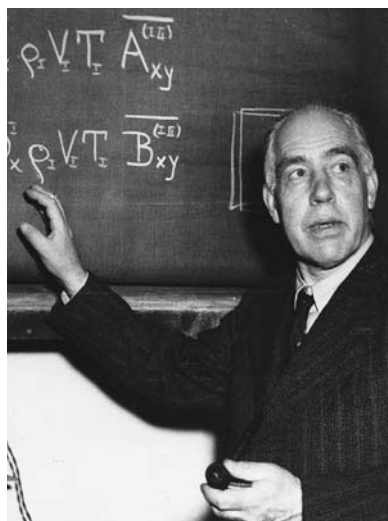


7.3

The Bohr Theory of the Hydrogen Atom

**FIGURE 7.8** ▲**Niels Bohr (1885–1962)**

After Bohr developed his quantum theory of the hydrogen atom, he used his ideas to explain the periodic behavior of the elements. Later, when the new quantum mechanics was discovered by Schrödinger and Heisenberg, Bohr spent much of his time developing its philosophical basis. He received the Nobel Prize in physics in 1922.

According to Rutherford's nuclear model, the atom consists of a nucleus with most of the mass of the atom and a positive charge, around which move enough electrons to make the atom electrically neutral. But this model, offered in 1911, posed a dilemma. Using the then-current theory, one could show that an electrically charged particle (such as an electron) that revolves around a center would continuously lose energy as electromagnetic radiation. As an electron in an atom lost energy, it would spiral into the nucleus (in about 10^{-10} s, according to available theory). The stability of the atom could not be explained.

A solution to this theoretical dilemma was found in 1913 by Niels Bohr (1885–1962), a Danish physicist, who at the time was working with Rutherford (see Figure 7.8). Using the work of Planck and Einstein, Bohr applied a new theory to the simplest atom, hydrogen. Before we look at Bohr's theory, we need to consider the line spectra of atoms, which we looked at briefly in the chapter opening.

Atomic Line Spectra

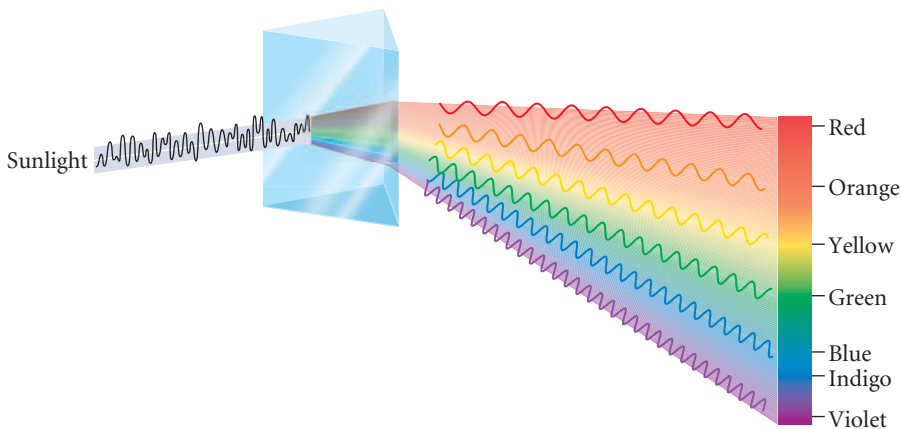
As described in the previous section, a heated solid emits light. A heated tungsten filament in an ordinary lightbulb is a typical example. With a prism we can spread out the light from a bulb to give a **continuous spectrum**—that is, a *spectrum containing light of all wavelengths*, like that of a rainbow (see Figure 7.9). The light emitted by a heated gas, however, yields different results. Rather than a continuous spectrum, with all colors of the rainbow, we obtain a **line spectrum**—a *spectrum showing only certain colors or specific wavelengths of light*. When the light from a hydrogen gas discharge tube (in which an electrical discharge heats hydrogen gas) is separated into its components by a prism, it gives a spectrum of lines, each line corresponding to light of a given wavelength. The light produced in the discharge tube is emitted by hydrogen atoms. Figure 7.2 shows the line spectrum of the hydrogen atom, as well as line spectra of other atoms.

The line spectrum of the hydrogen atom is especially simple. In the visible region, it consists of only four lines (a red, a blue-green, a blue, and a violet), although others appear in the infrared and ultraviolet regions. In 1885 J. J. Balmer showed that the wavelengths λ in the visible spectrum of hydrogen could be reproduced by a simple formula:

$$\frac{1}{\lambda} = 1.097 \times 10^7/\text{m} \left(\frac{1}{2^2} - \frac{1}{n^2} \right)$$

FIGURE 7.9 ►**Dispersion of white light by a prism**

White light, entering at the left, strikes a prism, which disperses the light into a continuous spectrum of wavelengths.



Here n is some whole number (integer) greater than 2. By substituting $n = 3$, for example, and calculating $1/\lambda$ and then λ , one finds $\lambda = 6.56 \times 10^{-7}$ m, or 656 nm, a wavelength corresponding to red light. The wavelengths of the other lines in the hydrogen atom visible spectrum are obtained by successively substituting $n = 4$, $n = 5$, and $n = 6$.

Bohr's Postulates

Bohr set down the following postulates to account for (1) the stability of the hydrogen atom (that the atom exists and its electron does not continuously radiate energy and spiral into the nucleus) and (2) the line spectrum of the atom.

1. Energy-level Postulate An electron can have only *specific energy values* in an atom, which are called its **energy levels**. Therefore, the atom itself can have only specific total energy values.

Bohr borrowed the idea of quantization of energy from Planck. Bohr, however, devised a rule for this quantization that could be applied to the motion of an electron in an atom. From this he derived the following formula for the energy levels of the electron in the hydrogen atom:

$$E = -\frac{R_H}{n^2} \quad n = 1, 2, 3, \dots \infty \quad (\text{for H atom})$$

The energies have negative values because the energy of the separated nucleus and electron is taken to be zero. As the nucleus and electron come together to form a stable state of the atom, energy is released and the energy becomes less than zero, or negative.

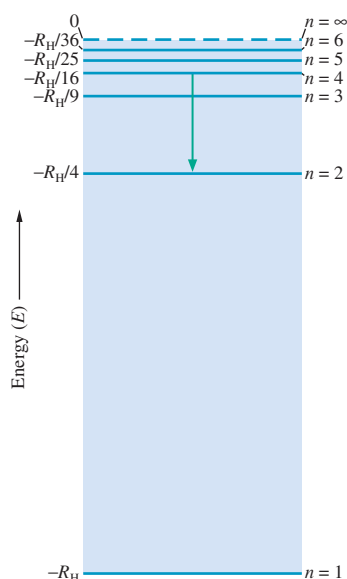


FIGURE 7.10 ▲
Energy-level diagram for the electron in the hydrogen atom

Energy is plotted on the vertical axis (in fractional multiples of R_H). The arrow represents an electron transition (discussed in Postulate 2) from level $n = 4$ to level $n = 2$. Light of wavelength 486 nm (blue-green) is emitted. (See Example 7.4 for the calculation of this wavelength.)

where R_H is a constant (expressed in energy units) with the value 2.179×10^{-18} J. Different values of the possible energies of the electron are obtained by putting in different values of n , which can have only the integral values 1, 2, 3, and so forth (up to infinity, ∞). Here n is called the *principal quantum number*. The diagram in Figure 7.10 shows the energy levels of the electron in the H atom. <

2. Transitions Between Energy Levels An electron in an atom can change energy only by going from one energy level to another energy level. By so doing, the electron undergoes a *transition*.

By using only these two postulates, Bohr was able to explain Balmer's formula for the wavelengths in the spectrum of the hydrogen atom. According to Bohr, the emission of light from an atom occurs as follows. An electron in a higher energy level (initial energy level, E_i) undergoes a transition to a lower energy level (final energy level, E_f) (see Figure 7.10). In this process, the electron loses energy, which is emitted as a photon. Balmer's formula follows by this reasoning: The energy lost by the hydrogen atom is $\Delta E = E_f - E_i$. If we write n_i for the principal quantum number of the initial energy level, and n_f for the principal quantum number of the final energy level, then from Postulate 1,

$$E_i = -\frac{R_H}{n_i^2} \quad \text{and} \quad E_f = -\frac{R_H}{n_f^2}$$

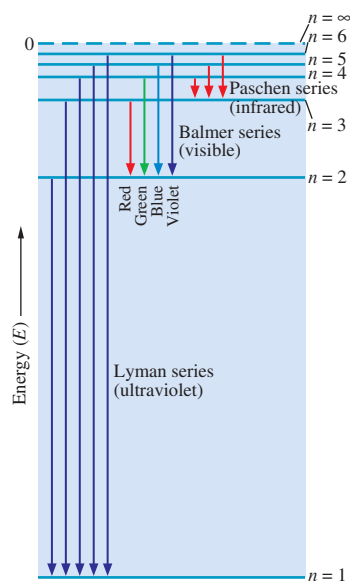
so

$$\Delta E = \left(-\frac{R_H}{n_f^2}\right) - \left(-\frac{R_H}{n_i^2}\right) = -R_H \left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$$

For example, if the electron undergoes a transition from $n_i = 4$ to $n_f = 2$,

$$\Delta E = -2.179 \times 10^{-18} \text{ J} \left(\frac{1}{2^2} - \frac{1}{4^2}\right) = -4.086 \times 10^{-19} \text{ J}$$

The sign of the energy change is negative. This means that energy equal to 4.086×10^{-19} J is lost by the atom in the form of a photon.

**FIGURE 7.11** ▲**Transitions of the electron in the hydrogen atom**

The diagram shows the Lyman, Balmer, and Paschen series of transitions that occur for $n_f = 1, 2$, and 3 , respectively.

In general, the energy of the emitted photon, $h\nu$, equals the positive energy lost by the atom ($-\Delta E$):

$$\text{Energy of emitted photon} = h\nu = -\Delta E = -(E_f - E_i)$$

That is,

$$h\nu = R_H \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$$

Recalling that $\nu = c/\lambda$, you can rewrite this as

$$\frac{1}{\lambda} = \frac{R_H}{hc} \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$$

By substituting $R_H = 2.179 \times 10^{-18} \text{ J}$, $h = 6.626 \times 10^{-34} \text{ J}\cdot\text{s}$, and $c = 2.998 \times 10^8 \text{ m/s}$, you find that $R_H/hc = 1.097 \times 10^7/\text{m}$, which is the constant given in the Balmer formula. In Balmer's formula, the quantum number n_f is 2. This means that Balmer's formula gives wavelengths that occur when electrons in H atoms undergo transitions from energy levels $n_i > 2$ to level $n_f = 2$. If you change n_f to other integers, you obtain different series of lines (or wavelengths) for the spectrum of the H atom (see Figure 7.11).

Example 7.4**Determining the Wavelength or Frequency of a Hydrogen Atom Transition**

What is the wavelength of light emitted when the electron in a hydrogen atom undergoes a transition from energy level $n = 4$ to level $n = 2$?

Problem Strategy Remember that the wavelength or frequency of a transition depends on the difference in energies of the levels involved. For an H atom, the energy levels are $E = -R_H/n^2$. You calculate the difference in energy for the two levels to *obtain the energy of the photon*. Then, calculate the frequency and wavelength of the emitted light. (Although you could do this problem by “plugging into” the previously derived equation for $1/\lambda$, the method followed here requires you to remember only key formulas, shown in color in the text.)

Solution From the formula for the energy levels, you know that

$$E_i = \frac{-R_H}{4^2} = \frac{-R_H}{16} \quad \text{and} \quad E_f = \frac{-R_H}{2^2} = \frac{-R_H}{4}$$

You subtract the lower value from the higher value, to get a positive result. (The energy of the photon is positive. If your result is negative, it means you calculated $E_f - E_i$, rather than $E_i - E_f$. Simply reverse the subtraction to obtain a positive result.) Because this result equals the energy of the photon, you equate it to $h\nu$:

$$\left(\frac{-R_H}{16} \right) - \left(\frac{-R_H}{4} \right) = \frac{-4R_H + 16R_H}{64} = \frac{-R_H + 4R_H}{16} = \frac{3R_H}{16} = h\nu$$

The frequency of the light emitted is

$$\nu = \frac{3R_H}{16h} = \frac{3}{16} \times \frac{2.179 \times 10^{-18} \text{ J}}{6.626 \times 10^{-34} \text{ J}\cdot\text{s}} = 6.17 \times 10^{14}/\text{s}$$

(continued)